

AATISH KUMAR

Senior Software Engineer | Full Stack & Generative AI

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PROFESSIONAL SUMMARY

Senior Software Engineer with **6+ years** building and shipping production-grade web applications and AI-powered products. Deep expertise in **full-stack architecture** — React, Next.js, TypeScript, Node.js, FastAPI, and PostgreSQL — combined with hands-on experience integrating **Generative AI** into real products: RAG pipelines, LLM orchestration (GPT-4, Claude, Gemini), and vector search (FAISS, Pinecone, ChromaDB). Comfortable owning end-to-end delivery: system design, cloud deployment on **AWS**, Docker-based CI/CD, automated testing, and production reliability. Not an AI researcher — an engineer who ships AI applications that work at scale. Experienced in designing and delivering production-grade AI-powered applications across the full software lifecycle, from architecture and deployment to monitoring and optimization.

TECHNICAL SKILLS

AI / GenAI: OpenAI GPT-4 · Claude API · Gemini API · RAG Pipelines · LLM Orchestration · Prompt Engineering · Vector Search · MCP Server Design · Tool Calling

Frontend: React.js · Next.js · TypeScript · Redux · Context API · Tailwind CSS · Material UI · SPA / PWA

Backend: Node.js · Express.js · FastAPI · Python · REST API Design · Redis · Background Jobs · API Caching

Databases: PostgreSQL · MongoDB · MySQL · FAISS · Pinecone · ChromaDB · Schema Design · Query Optimisation

Cloud & DevOps: AWS (EC2 · S3 · Lambda · Inspector) · Docker · CI/CD Pipelines · Git · Infrastructure Rightsizing

Testing & Quality: Jest · Mocha · Chai · SonarQube · Test Automation · Security Scanning · Vulnerability Remediation

PROFESSIONAL EXPERIENCE

Senior Software Engineer **LeewayHertz (A Hackett Group Company)** Dec 2024 – Present | Remote

ZBrain AI Platform · Quantadoc · HRJD Builder

- Architected and delivered **RAG-based document intelligence workflows** on the ZBrain AI platform — integrating GPT-4, Claude, and Gemini APIs into production **FastAPI** services handling enterprise document ingestion, semantic chunking, embedding indexing, and LLM response generation.
- Designed and maintained **FAISS and Pinecone vector index pipelines** for embedding storage and similarity retrieval — powering semantic search across large enterprise document corpora.
- Engineered **LLM orchestration and prompt engineering workflows** for the HRJD Builder — automating structured HR job description generation, reducing manual documentation effort for HR teams.
- **Optimised API response pipelines and vector retrieval queries** — achieving a **15–20% reduction in end-to-end latency** across production AI workflows.
- Delivered production **React frontends** for AI-powered document interaction and LLM workflow interfaces — translating complex backend AI logic into responsive, user-facing product experiences.
- Managed **Docker-based deployments and CI/CD pipelines**; resolved AWS Inspector security findings and rightsized EC2/Lambda resources — improving security posture and reducing infrastructure costs.
- Established **Jest, Mocha, and Chai** test coverage for shared AI service modules; integrated **SonarQube** for continuous code quality and security scanning across the ZBrain platform.

Full Stack Developer **Appinventiv** Apr 2022 – Sep 2024 | Chandigarh

End-to-end delivery across 5+ enterprise web products

- Owned full technical delivery of **5+ enterprise web applications** — leading architecture, development, and production deployment across **React, Next.js, TypeScript, Node.js, Express.js, PostgreSQL, and MongoDB**.
- Engineered high-performance **React / TypeScript frontends** with component-level rendering optimisations — achieving **~30% reduction in page load times** across apps serving **10,000–50,000 monthly active users**.
- Designed **scalable REST API architectures and PostgreSQL database schemas** from inception on 3+ greenfield projects — covering data modelling, endpoint design, validation logic, API versioning, and query optimisation.
- Deployed applications on **AWS (EC2, S3)** using Docker containers with CI/CD pipeline configuration — ensuring consistent, zero-downtime production releases.

- Built **reusable React component libraries** and shared design system patterns — measurably accelerating UI delivery velocity across parallel product teams.
- Enforced **SonarQube quality gates**, maintained automated test suites, and conducted regular code reviews — contributing to significantly lower post-release defect rates.

Full Stack Developer **Digimantra Labs** Jan 2020 – Mar 2022 | Mohali, Punjab

- Led **end-to-end delivery of 4+ client projects** — owning architecture decisions, stakeholder communication, AWS deployment, and cross-functional execution across concurrent engagements.
- Built responsive **React frontends backed by Node.js RESTful APIs** — achieving **~20% reduction in page render time** through targeted component-level optimisation.
- Mentored junior engineers, led code reviews, and enforced testing standards — raising team engineering quality across active projects.
- **Awarded Star Performer (2021–2022)** among 30+ engineers for consistent on-time delivery and technical quality across multiple simultaneous client engagements.

Trainee Web Developer **Innow8 Apps** May 2019 – Dec 2019 | Mohali, Punjab

- Shipped 10+ REST API endpoints in Node.js / MongoDB and built React / Redux UI modules — delivered an Admin Portal that **improved internal productivity by ~40%** through streamlined UX and optimised data workflows.

KEY PROJECTS

MCP Server — Agentic AI & Tool Calling · Personal Engineering Project · GitHub

- Built a production-ready **Model Context Protocol (MCP) server** from scratch — implementing tool registration, request routing, and protocol handling to enable **LLM agents** to discover and invoke external tools dynamically. Hands-on experience with **agentic AI architecture**, **tool calling**, and multi-step AI workflow design — directly applicable to building autonomous AI agents and LLM-powered automation systems.

Quantadoc — Full Stack Document AI Platform · LeewayHertz

- Delivered a production **full-stack document AI platform** end-to-end — engineered FastAPI semantic retrieval backend with FAISS / Pinecone embedding pipelines and built the React frontend for interactive AI document workflows, from ingestion through LLM-generated outputs.

ZBrain AI Platform — Enterprise RAG System · LeewayHertz

- Contributed to enterprise **RAG pipeline development** and multi-model LLM orchestration (GPT-4, Claude, Gemini) — FastAPI backend services supporting large-scale document ingestion, vector indexing, and semantic retrieval for enterprise clients.

EDUCATION

B.Tech – Computer Science Indira Gandhi Institute of Technology and Management, Sohna · 2016–2019

Diploma – Computer Science & Engineering Govt. Polytechnic Hamirpur · 2013–2016